POSTULATE 4

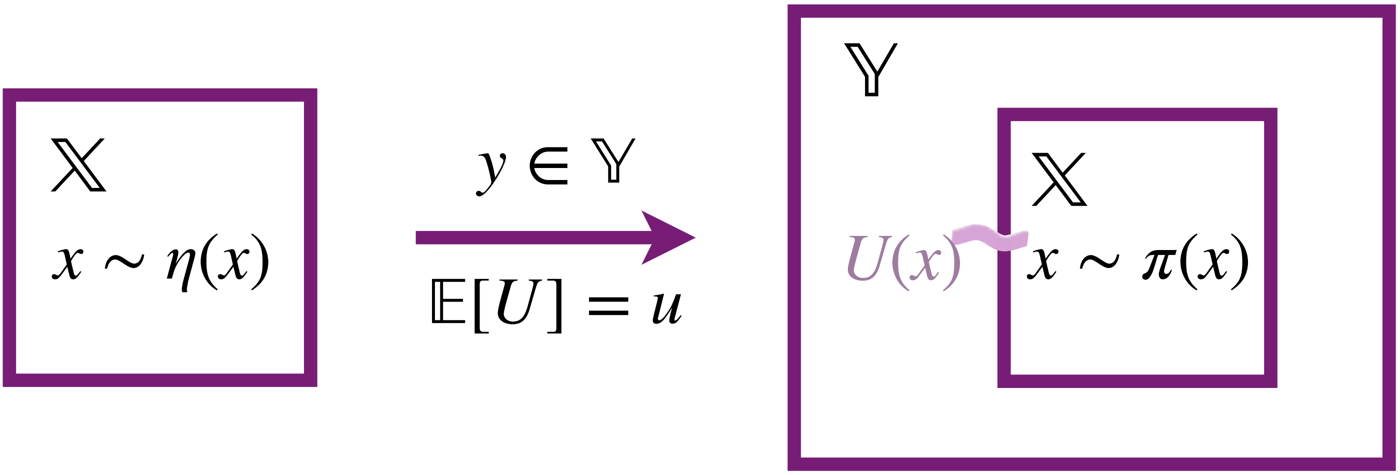
8

- **Postulate 4:** Given the reference and the energy constraint, the equilibrium distribution π is the one that is **as close to η as possible** while satisfying the constraint.

- Closeness is measured by KL divergence:

$$\text{minimize}: \qquad \mathrm{KL}(\pi \| \eta) = \pi\left[\log \frac{\mathrm{d}\pi}{\mathrm{d}\eta}\right]$$

$$\text{Constraint}: \qquad \pi[1] = 1, \qquad \pi[U] = u$$



$\mathbb{X}$
$x \sim \eta(x)$
$y \in \mathbb{Y}$
$\mathbb{E}[U] = u$
$\mathbb{Y}$
$\mathbb{X}$
$U(x)$
$x \sim \pi(x)$

- Postulte 4 is Jaynes' maximum entropy principle: “don’t assume anything beyond the constraint”

- We define the relative entropy of π with respect to η as

$$S(\pi) = -\operatorname{KL}(\pi \| \eta)$$

$$= - k_B \mathrm{KL}(\pi \| \eta)$$

- In chemistry we introduce the Boltzmann constant $k_B = 1.380649 \times 10^{-23}$ joules per Kelvin

POSTULATE 4

- **Postulate 4:** Given the reference and the energy constraint, the equilibrium distribution π is the one that is **as close to η as possible** while satisfying the constraint.
- Closeness is measured by KL divergence:

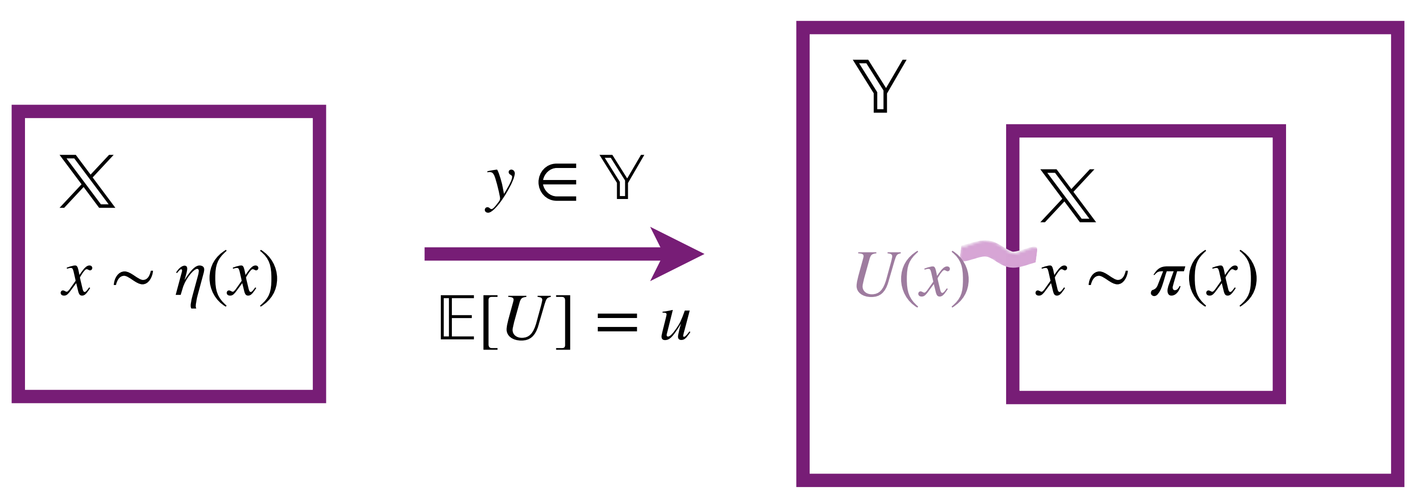
$$\text{minimize}: \qquad \mathrm{KL}(\pi \| \eta) = \pi\left[\log \frac{\mathrm{d}\pi}{\mathrm{d}\eta}\right]$$

$$\text{Constraint}: \qquad \pi[1] = 1, \qquad \pi[U] = u$$

- We define the relative entropy of π with respect to η as

$$S(\pi) = -\,\mathrm{KL}(\pi \| \eta) \;\; = -\,k_B \mathrm{KL}(\pi \| \eta)$$

  - In chemistry we introduce the Boltzmann constant $k_B = 1.380649 \times 10^{-23}$ joules per Kelvin
- Postulte 4 is Jaynes' maximum entropy principle: “don’t assume anything beyond the constraint”



ENTROPY